PAVAN KUMAR GUDLA

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PROFESSIONAL SUMMARY

Applied AI/ML Product Engineer with 2+ years of software engineering experience building enterprise GenAI applications and agentic workflows. Experienced in Retrieval-Augmented Generation (RAG), hybrid retrieval, knowledge graphs, and LLM orchestration using Python, LangGraph, LangChain, and Neo4j. Contributed to AI-powered application engineering workflows spanning specification analysis, frontend code generation, API integration, and LLM observability. Currently pursuing an M. Tech in Artificial Intelligence and Machine Learning at BITS Pilani.

WORK EXPERIENCE

Applied AI/ML Product Engineer – Agentic AI & GenAI | Tata Consultancy Services, Hyderabad | *Mar 2024 – Present*

- Translated evolving business requirements into AI solution designs and working GenAI/Agentic AI workflows, collaborating with product and engineering stakeholders through iterative development, testing, and product integration while emphasizing validation, traceability, and workflow observability.
- Designed and developed multi-agent workflows for an enterprise application-generation platform, transforming business and UI specifications into structured application models, UI previews, and generated code, with a reported reduction of approximately 90% in developer effort.
- Developed GenAI and Agentic AI PoCs for enterprise application engineering, translating evolving requirements into technical designs and working AI workflows through iterative experimentation, testing, and integration.
- Built a Context Analyzer Agent that extracts intent, requirements, screens, fields, widgets, events, and actions from structured and unstructured inputs (BRDs, wireframes, images, HTML), replacing manual requirement analysis 100%.
- Integrated AI agents with Neo4j through API-based JSON model population, storing structured application specifications and meta-model data for retrieval by downstream agents during application-generation workflows.
- Developed Screen Preview and Code Generator Agents that retrieve specs from Neo4j to generate HTML previews and frontend code, shipped in TCS MasterCraft v9.0.0 production release.
- Built and integrated Retrieval-Augmented Generation (RAG) pipelines using text-embedding models, document vectorization, vector search, semantic search and multiple retrieval methods to provide relevant document context to LLM-powered agent workflows.
- Implemented hybrid retrieval capabilities combining embedding-based semantic search and BM25 keyword retrieval, alongside Neo4j Knowledge Graphs and Cypher queries, to retrieve relevant information from document collections and structured enterprise application data.
- Developed reusable Python tools for document and HTML parsing, JSON generation, knowledge retrieval, validation, image preprocessing, and code generation. Integrated internal tools and services through the Model Context Protocol (MCP).
- Integrated enterprise LLMs, including GPT-4.1 Mini and GPT-OSS 120B, and configured system prompts, temperature, top-k, and top-p parameters to support structured, context-grounded generation across AI workflows.
- Implemented LLM workflow observability using Opik to track token consumption, model interactions, agent execution, and workflow-level performance. Contributed to agent architecture design, debugging, and unit, integration, and performance testing within an Agile development lifecycle.

EDUCATION

M. Tech in Artificial Intelligence and Machine Learning (WILP) | Birla Institute of Technology and Science, Pilani (BITS Pilani) | Oct 2025 – Sep 2027

B. Tech in Mechanical Engineering | Jawaharlal Nehru Technological University, Kakinada (JNTUK) | Aug 2019 – Jun 2023

TECHNICAL SKILLS

Programming & Software Engineering:

Python • FastAPI • REST APIs • Git • API Integration • Debugging • Unit Testing • Integration Testing • Validation Pipelines

Frontend & Application Development:

HTML • Next.js • React • TypeScript • Streamlit

AI/ML & Generative AI:

Generative AI • Large Language Models (LLMs) • Agentic AI • Multi-Agent Systems • Agent Orchestration • Tool Calling • Function Calling • Structured Outputs • Prompt Engineering • Context Engineering

LLM Frameworks & Platforms:

LangGraph • LangChain • CrewAI • Azure OpenAI • OpenAI • OpenRouter • MCP (Model Context Protocol)

RAG & Knowledge Systems:

Retrieval-Augmented Generation (RAG) • Text Embeddings • Semantic Search • Vector Search • ChromaDB • pgvector • Hybrid Retrieval • BM25 • Neo4j • Knowledge Graphs • Cypher • Contextual Grounding

AI Engineering & Observability:

LLM Observability • Agent Tracing • Token Usage Monitoring • Opik • Model Evaluation • Output Validation • Human-in-the-Loop (HITL)

Machine Learning Fundamentals:

Machine Learning • Deep Learning • Natural Language Processing (NLP) • Recommendation Systems • Probability & Statistics • Linear Algebra

Cloud, Deployment & DevOps:

Microsoft Azure • AWS • Docker • Kubernetes • CI/CD • Vercel

PROJECTS

IncidentPilot AI | Multi-Agent Incident Analysis Platform

- Built an AI-powered incident analysis workflow using Python and LangGraph, with specialized triage, diagnosis, and resolution stages.
- Developed a FastAPI backend and Next.js frontend to connect incident inputs with structured analysis results through an API.
- Added automated tests for backend health and analysis workflows; persistence and live agent streaming remain planned for later iterations.

RecoMart | Recommendation & Data Management Pipeline

- Built an end-to-end recommendation pipeline over 100,836 ratings and 9,742 movies, combining ratings data with movie and TMDB metadata.
- Implemented data ingestion, validation, cleaning, feature engineering, dataset construction, model training, and recommendation generation.
- Constructed collaborative-filtering, content-based, and hybrid datasets; trained an SVD collaborative-filtering model and evaluated results using RMSE and MAE.

Autonomous Research Scout | Research Intelligence Platform

- Built a research discovery and analysis workflow gathering AI research from arXiv, GitHub, and Papers with Code.
- Designed retrieval, LLM-based analysis, prioritization, trend detection, and personalized research recommendation stages.
- Developed and deployed v1 an interactive application presenting AI-generated summaries, research trends, repositories, and personalized reading plans on Streamlit.

FraudGuard AI | PyTorch-Based Fraud Detection & AWS ML Serving

- Building a deep-learning fraud detection system using PyTorch, comparing Multilayer Perceptron (MLP) and autoencoder models against conventional machine learning baselines on an imbalanced transaction dataset.
- Developing an ML evaluation pipeline using PR-AUC, precision, recall, F1-score, and cost-sensitive threshold selection to assess fraud detection performance.
- Designing a cloud-native inference architecture using FastAPI, Docker, Amazon S3, Amazon ECR, and Amazon ECS Fargate, with CI/CD automation and CloudWatch monitoring.

CERTIFICATIONS

- Claude Certified Architect Professional
- Microsoft Certified: Azure AI Engineer Associate
- Microsoft Certified: Azure Data Scientist Associate
- Neo4j Certified Professional